

ABRAR ZAHIN RAIHAN

✉ Dhaka, Bangladesh

🌐 [azraihaan.github.io](https://github.com/azraihaan)

✉ zaheenraian@gmail.com

🌐 [Abrar Zahin Raihan](#)

🌐 [azraihaan](#)

EDUCATION

Bangladesh University of Engineering and Technology

Bachelor of Engineering in Computer Science and Engineering

Nov 2022 – Present

Dhaka, Bangladesh

Notre Dame College

College

Jul 2019– Dec 2021

Dhaka, Bangladesh

St. Joseph Higher Secondary School

High School

Jan 2016– Feb 2019

Dhaka, Bangladesh

INTERESTS

NLP · Information Retrieval · Multimodal AI · AI Systems · Computer Vision · Machine Learning · Mechanistic Interpretability

PUBLICATIONS

📄 **Causal Localization of the English Pivot in LLaVA: Mechanistic VLM Analysis and Training-Free Multilingual Steering** **2026**

1st Workshop on Multilinguality in the Era of Large Language Models (MeLLM) @ ACL 2026

San Diego, CA

Accepted · Archival (To appear in ACL Anthology)

- Causally localized an English-pivot bottleneck in LLaVA-1.5-7B to layers 5–17 via logit-lens analysis and causal activation patching; training-free steering vectors improved Russian VQA by +6.5 pp and Portuguese by +4.0 pp on MMB without fine-tuning.
- [🔗 Code](#)

📄 **Where Do Affordances Crystallize? Video vs. Image Self-Supervised Encoders for Embodied Perception** **2026**

7th Annual Embodied AI (EAI) Workshop @ CVPR 2026

Denver, CO

Accepted · Non-archival · Poster (To appear on workshop website)

- Probed 25 layer positions of ViT-L encoders (V-JEPA 2/2.1, DINOv2); video SSL crystallizes affordance signal 4 layers earlier than image SSL; top-3 attention heads achieve 5.3× compute reduction with ≤ 0.1 mAP loss.
- [🔗 Code](#)

📄 **SpatialCoT: Chain-of-Thought Spatial Grounding for Viewpoint-Robust 3D-LLMs** **2026**

2nd Workshop on 3D-LLM/VLA: Bridging Language, Vision and Action in 3D Environments @ CVPR 2026

Denver, CO

Under Review

- SpatialCoT: five-step geometry-grounded CoT (Localize, Orient, Enumerate, Compute, Answer) that makes 3D-LLMs robust to viewpoint rotation; reduces accuracy drop at 180° rotation from 77% to 41% on Real-3DQA.

EXPERIENCE

🌐 **[SysModeler.ai](#)**

November 2025 – February 2026

Research Assistant (Remote)

Canton, Michigan, United States

- **SysML Layout Algorithms** Designed and implemented layout algorithms to generate structured, readable, and visually consistent SysML diagrams
- **Retrieval-Augmented Image-to-SysML** Developed a **retrieval-augmented** image-to-SysML pipeline — given an input image, the system **retrieves** structurally similar exemplars as anchors to guide generation of semantically accurate SysML representations; grounding **multimodal output** in **retrieved references** to enforce structural consistency

🌐 **[Syslabo Corp](#)**

Jul 2024 – Sep 2024

SWE Intern (Remote)

Tokyo, Japan

- **Org Chart Automation** Automated creation and tracking of organizational hierarchy charts, replacing a manual Excel-based system with a full-stack MERN application
- **System Design** Designed scalable database schemas and RESTful APIs for efficient hierarchical data management
- **Outstanding Performance Recognition** Received best architecture design recognition among all interns; secured a full-time job offer
- [🔗 Internship Project](#)

PROJECTS

- 🌐 [IEEE VIP Cup 2025](#) | *Multimodal AI · Computer Vision · AI Systems* **Jul 2025**
 - **Computer vision** and **multimodal AI** system — **cross-modal RGB-IR fusion** for real-time drone detection, tracking, and payload classification under real-world degradations (blur, noise, camera instability); Champions among 40+ international teams, presented at ICIP 2025, Anchorage, AK.
 - 📄 [Presentation](#) 📄 [Paper](#) 🔗 [Code](#)
 - 🌐 [Read more](#)
- 🔗 [Intent Based Multilingual Search Optimization](#) | *NLP · Information Retrieval · AI Systems* **Apr 2025**
 - NLP-driven **information retrieval** system — **intent decomposition** and **multilingual dense retrieval** with in-memory **LLM inference** via llama's C++ runtime and 4-bit quantization, running CPU-only on constrained hardware; Champions among 86 teams at the Poridhi.io × Brain Station 23 AI Engineering Hackathon.
 - 📄 [Presentation](#) 🔗 [Code](#)
 - 🌐 [Read more](#)
- 🌐 [ASR for Low Resource Bengali Regional Dialect](#) | *NLP · Multimodal AI · Machine Learning* **Nov 2025**
 - **Multilingual NLP** and **multimodal AI** task — **low-resource ASR** bridging **acoustic and text modalities** across 20 regional Bangladeshi dialects to standard Bangla via fine-tuned Whisper ensembles and **LLM-augmented** data synthesis; 1st Runner-Up among 90+ teams at CUET ETE Televerse 1.0.
 - 📄 [Presentation](#) 🔗 [Code](#)
 - 🌐 [Read more](#)

HONORS & AWARDS

- 🌐 [CUET ETE Televerse 1.0 : AIfication 2025 \(Datathon\) - First Runner-up](#) Low-resource multilingual NLP — bridging acoustic and text modalities across 20 Bangladeshi dialects to standard Bangla via fine-tuned Whisper ensembles and LLM-augmented data synthesis. **Nov 2025**
- 🌐 [IEEE VIP Cup 2025 - Champions](#) Multimodal AI and computer vision — cross-modal RGB-IR fusion for real-time UAV detection, tracking, and payload classification; end-to-end AI system presented at ICIP 2025, Anchorage, AK. **Jul 2025**
- 🌐 [Poridhi.io and Brain Station 23 AI Engineering Hackathon - Champions](#) NLP-driven information retrieval — intent decomposition and multilingual dense retrieval with in-memory LLM inference via llama's C++ runtime, CPU-only on constrained hardware. **Apr 2025**
- 🌐 [BUET CSE Fest Hackathon 2024 - 2nd Runner-up](#) Cloud-native systems engineering — microservices on Azure Kubernetes Service with Redis caching, automated CI/CD, and full Prometheus/Grafana observability. **Oct 2024**
- 🌐 [Therap JavaFest 2024 - Top 10 finalist](#) RAG-powered conversational assistant, semantic retrieval-based project recommender, and computer vision sketch-to-image pipeline — gamified coding platform for children; top 10 among 1300+ participants. **Sep 2024**
- 🌐 [Code Samurai Hackathon 2024 - 10th Place](#) Engineered a waste management system for the Dhaka North City Corporation (DNCC) (as proposed by BJIT), solved algorithmic challenges, and built a web app and a React Native mobile app to implement the system **Mar 2024**